

Project Tricorder Mk I · Field manual · Rev 0.6.92

Jim Ames, Claude and Grok · N2NHU Labs · Newburgh, New York 12550. RENT-A-HAL Android Edition. Supersedes 0.6.80. Companion on-device file: ui-shell/manual.html. This load is 0.6.92.

NAVIGATION WARNING. Not certified by FAA, NASA, EASA, or ICAO. The cover-cockpit ADI, heading rose, G-meter, ASI-style GPS speed dial, barometric ALT tape, VSI tape, pressure-altitude / QNH readouts, satellites, stars, SOL orbits, moon phase, sunrise/sunset, moonrise/moonset, GNSS clock, NAV dead-reckoning, and HF forecasts **MUST NEVER** be used to fly, land, taxi, dock, or navigate any craft. Sun and moon times are field approximations (± 3 min on the moon), not an almanac. The satellite clock is a field novelty, not a time standard. This load is not DO-178C airborne software. There is no pitot. “ASI” is groundspeed, not indicated airspeed.

WATCH WARNING. Weapon and gunfire lines are caption keywords and YAMNet labels. Not a weapons detector. Not a 911 substitute.

MEDICAL NOTE. MEDICAL is an encyclopedia briefing, not a clinician. MEMORY ALPHA is Simple English Wikipedia, not the Star Trek wiki.

MARKS. Not authorized by or affiliated with The Wand Company, Paramount SkyDance, Desilu, Gulf and Western, CBS, Roddenberry Entertainment, or Wah Chang.

What 0.6.81–0.6.92 added

This block is the delta since Field Manual 0.6.80. The icon face, Audio books, NAV callsigns, SOL luna, Comlink, dictation, flights, and Termux four-pack are unchanged in spirit; see the later recap.

0.6.81–0.6.84 — landscape instrument cluster. Closed Fold, Horizon on, rotate to landscape: the cover is an edge-to-edge ADI. Right rail is ASI (circular GPS-speed dial with mph / km/h / knots / Mach annotations) · G tape · ALT tape · VSI tape. Labels sit inside the rails. Portrait strip is untouched.

0.6.85–0.6.86 — heading false starts. Display-matrix remap and a pitch-hold on the rose were tried and rejected. They swapped east/west or froze the rose. 0.6.86 restored raw azimuth and dropped the latch.

0.6.87–0.6.88 — PFD tapes + a syntax outage. ALT became a moving scale with a fixed mid-window (G1000 style). VSI first used a fixed scale and a caret. 0.6.87 shipped a duplicate *const a0* in the same tick and the face went blank. 0.6.88 renamed the iterator.

0.6.89 — world-up VSI and a live G meter. G-load is |accelerometer| / g, not |gravity| / g (the latter is always ~ 1). VSI briefly blended GPS height-rate and gravity-axis IMU. GPS height jitter made that a bad idea.

0.6.90 — windscreen frame. Hold the closed unit in front of your eyes, long edge level, as if it were a glareshield — not flat on a desk. Heading is the look axis through the glass (device –Z projected on the horizon): the direction you are facing. ADI pitch/roll come from gravity remapped so screen-up is sky. Table-top “compass on a picnic table” axes are gone.

0.6.91–0.6.92 — baro owns the tapes. QNH is a one-shot snapshot from the first sane GPS height + station pressure, then frozen. MSL ALT and the ALT tape/drum are station baro against that hold (or ISA pressure altitude if no hold yet). VSI is $d(\text{baro height})/dt$ only. GPS ALT remains a text line with a Δ versus baro. It does not drive the instruments.

Cover cockpit — how to stand

Arm Horizon. Fold closed. Rotate to landscape. Face hides; cover is the PFD. Rotate back to portrait for the tricorder face. If Horizon is off, landscape is the ordinary narrow layout.

Hold it in your line of sight, screen toward you, world behind the phone — the same pose as looking through a windscreen. Do not lay it on a table and expect the rose and brown to agree with “north out the windshield.” Flat on a desk, the look axis points at the floor.

Aircraft symbol is fixed. Sky/ground follow the real horizon (Western ADI). Nod: brown climbs when the nose goes down. Bank: the horizon rolls. G tape 0–3 with a 1 G mark; bounce the unit and it should leave 1.0.

The pressure lines (what the letters mean)

Line	Meaning
STN hPa abs	Station pressure. Raw barometer at your hand. Not sea-level. Your ~994 hPa on a 1015 hPa map was this.
MSL hPa QNH hold	Estimated sea-level pressure, computed once from STN + first GPS height, then frozen so GPS jitter cannot chase it.
PA ... ISA 1013.25	Pressure altitude. Textbook atmosphere with 1013.25 hPa at sea level. Pilots get this on 29.92 in. Not AGL.
MSL ALT ... baro	Height from STN using the held QNH. After the hold, this is a baro altimeter. Stairs move it. GPS twitch does not.
GPS ALT ... Δ	GNSS height only. Δ is baro MSL minus GPS. Tens of metres is normal. Not an instrument source.
ASI dial	GPS groundspeed with unit bands. Not indicated airspeed. There is no pitot, no ram air.
VSI tape	Sliding scale, fixed mid-window, d(baro height)/dt. Climb numbers ride up.

Why aircraft grow tubes: the Fold measures cabin air in a sealed slab. A ship pipes **static** (flush port) to alt/VSI and **pitot** (ram) minus static to ASI. This unit has only static-class baro plus GNSS. Fun on a stair or a fair-ride G tape. Not a pitot-static system.

Face (unchanged since 0.6.80)

Phosphor line icons. Long-press speaks the name. Inquiry seated at a desk. Sensor plate + ECG + pings. Sentry cap. Audio ear (off → green listen → blue planes book → yellow alerts book → off). Translator two marks. Horizon/NAV outline delta in a ring. Satellites/stars field. Sol sunburst (space → earth weather → moon → sunrise/sunset → luna rise/set → off). Logs: book, pen, talking head, play-on-book. IPL/ALL abort. STT kills speech only.

ROBO four-state: dark robot = Kokoro · green robot = Android TTS · yellow one-ear headset = robot + hands-free TRICORDER menu · red mute bar = silent observer (CRT and log only). Next touch returns Kokoro.

Recap still in force

Audio books (0.6.77). Green listen, blue planes carousel (last 80 cards), yellow civic/siren book, then off. Books do not steal each other.

NAV (0.6.42–0.6.77). Walking dead-reckon. CAMP, MARK, DEST as lat/lon with N/S E/W, plus code, or W3W. North-up sketch. Named Comlink units on the map. Off-track chime ~30 m. Not for driving or flight.

SOL / sky. Space weather + HF/VHF/UHF, local forecast, moon phase, sunrise/sunset (NOAA eq., local noon), moonrise/moonset (Meeus-lite, ±3 min, rise/set azimuth). Star-field key toggles birds and stars.

Comlink. Friendly name persists. Nearby + optional Meshtastic/LoRa. Backup/restore v2 zip to Downloads/ProjectTricorder and app files. Mesh GPS every 60 s.

Flights. OpenSky geobox (settings). Tail, type, photo, airframe card, FROM/TO city dossier, AUTOPLAY. English/metric toggle. Robot voice for alerts. Rate-limit 429 is spoken if not MUTE.

Watch. Sentry captions every dwell; weapon/possible-weapon tones. Audio: gunfire / scream / siren / traffic classes with fuzzy quieting on some false friends. Citizen cards when enabled.

Dictation. Wake DICTATION. NEW / EDIT / LOG / SUMMARIZE / MINDMAP / SPEAK / KILL. ZULU SAVE or ABORT. Cursor, punctuation, CONFIRMED deletes.

Banks. MEMORY ALPHA (Simple English Wikipedia), MEDICAL (WikiMed titles + Wikipedia body). On-device YAMNet; optional BirdNET and flora/fauna fetches. Translator packs.

Backends. Termux four-pack: chat :8080, vision :8081, image :8188, tts :5000. `~/bin/tricorder start`. Imagine sliders: size 256/384/512 and steps. See TERMUX-BACKEND-INSTALL.md.

Troubleshooting the glass

Blank face after an install — usually a script syntax error (0.6.87). Reinstall 0.6.88+.

Rose correct only when the phone is flat — you are in table-top pose. Hold it as a windscreen (0.6.90+).

G tape stuck at 1.00 — pre-0.6.89 used the gravity sensor. This load uses the accelerometer.

MSL ALT glued to GPS ALT — pre-0.6.92 recomputed QNH every frame from GPS. This load freezes QNH.

VSI twitchy — older blends used GPS height-rate. 0.6.92 is baro only.

Horizon brown on top while standing — confirm Horizon is on, unit closed, landscape, screen toward your face, long edge level.

Still a toy. Kelly Johnson would stamp the bezel. So do we.